

Lecture 8: Security Aspects of the File System

- File-system security controls who can access persistent data and what operations they can perform.
- Key objectives: confidentiality, integrity and availability.
- Security mechanisms must balance protection, usability and performance.

Learning Objectives

- Distinguish authentication from authorization.
- Explain ownership and UNIX permission bits.
- Explain ACL-based access control.
- Describe encryption, auditing and secure deletion.
- Analyze common file-system security threats.
- Evaluate security-performance trade-offs.

Security Objectives

Confidentiality

Prevent unauthorized disclosure.

Integrity

Prevent unauthorized modification.

Availability

Ensure authorized access remains possible.

File-system controls contribute to all three objectives.

Authentication vs. Authorization

- **Authentication:** establishes the identity of a user/process.
- **Authorization:** determines what that identity is permitted to do.

Who are you? → Authentication

What may you do? → Authorization

Authorization can be implemented through permissions, ACLs and other access-control mechanisms.

UNIX Permission Model

`-rwxr-x---`

Owner	Group	Others
<code>rwX</code>	<code>r-X</code>	<code>---</code>

- `r`: read
- `w`: write
- `x`: execute for files; search/traverse permission for directories

Numeric Permissions

Permission	Value
read	4
write	2
execute	1

Example:

$$rwx = 4 + 2 + 1 = 7, \quad r-x = 4 + 0 + 1 = 5$$

Therefore:

$$rwxr-x--- = 750$$

Directory Permissions

- **Read:** ability to list directory entries.
- **Write:** ability to create/delete/rename entries, subject to other controls.
- **Execute/search:** ability to traverse the directory and access known entries.

Important

File permission and directory permission are separate. A user may have permission on a file but still be unable to reach it if directory traversal is denied.

Access Control Lists (ACLs)

- Traditional owner/group/other permissions may be too coarse.
- ACLs allow permissions to be assigned to specific users or groups.

Professor	read/write
Student-A	read
Student-B	read
Other	no access

ACL evaluation rules depend on the operating system and ACL model.

- **File-level encryption:** protects selected files.
- **File-system/storage encryption:** protects broader storage areas.
- **Full-disk/device encryption:** protects data on a lost/stolen device.
- Encryption protects confidentiality, but key management is critical.
- Encryption can add CPU, I/O and latency overhead, although modern hardware may reduce the cost.

Security Threats

- Unauthorized file access
- Privilege escalation
- Malicious modification
- Malware and ransomware
- Data leakage
- Insecure temporary files
- Symbolic-link and path-manipulation attacks
- Recovery of supposedly deleted data

Auditing and Secure Deletion

- **Auditing**: record security-relevant events such as access failures and sensitive operations.
- Logs can support detection, investigation and accountability.
- **Secure deletion**: depends on storage technology.
- On flash storage, wear leveling and remapping can make traditional overwrite assumptions unreliable.
- Encryption plus secure key destruction can provide a practical data-disposal strategy in some systems.

Security–Performance Trade-Off

- Stronger access checks can increase CPU or metadata overhead.
- Encryption can introduce computation and I/O overhead.
- Auditing increases storage and processing activity.
- Caching improves performance but must not bypass authorization semantics.

Security $\leftrightarrow$ Performance $\leftrightarrow$ Usability

University Research Server

A research directory contains confidential datasets. Faculty require read/write access; students require read-only access; external users must have no access.

Ask students to:

- 1 define users/groups,
- 2 propose permissions/ACLs,
- 3 identify encryption requirements,
- 4 identify audit events,
- 5 discuss performance implications.

- **Galvin, 10th Ed.:** Chapter 16 – Security; Chapter 17 – Protection, including access control and UNIX-style protection concepts.
- **Stallings, 9th Ed.:** Chapter 21 – Operating System Security; access control, Unix access control, hardening and security maintenance.

- File-system security combines authentication, authorization and protection mechanisms.
- UNIX permissions provide a compact basic access-control model.
- ACLs support more fine-grained policies.
- Encryption protects data confidentiality; auditing supports accountability.
- Security mechanisms must be evaluated together with their operational and performance costs.